CONFIRM: Cross-Checking Observed NIRSpec Fits for Improved Redshift Measurements

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PRISMATIC
PROBING REDSHIFTS AND INVESTIGATING SPECTRA OF MULTIPLE ASTRONOMICAL TARGETS IN THE CANDELS-AREA

Introduction

As the universe expands, galaxies move away from each other, causing their emitted light to stretch to longer wavelengths in a phenomenon known as redshift. The greater a galaxy's redshift, the farther away it is, making redshift measurements essential for studying the early universe. The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS) uses JWST's NIRSpec/PRISM instrument to observe candidate high-redshift galaxies.[1] Determining accurate redshifts requires extracting spectral features from emission spectra, often done by using automated fitting codes such as BAGPIPES, msaexp, and CIGALE. To ensure the reliability of these automated measurements, human verification is not only necessary but also vital to our goal of creating a definitive redshift catalog.

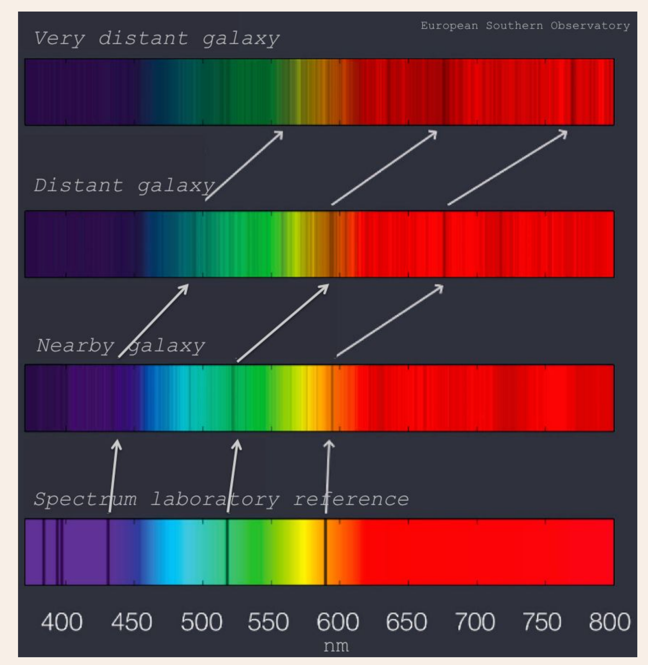

Figure 1. Redshift as a function of distance.[2]

Research Goal

By systematically comparing 1,039 automated redshift sources to independently calculated values, CONFIRM improves the accuracy of galaxy classification and provides insight into the strengths and limitations of the fitting codes used in PRISMATIC. Our team then uses this training to classify an incoming dataset of 8,000 sources to continue our human-verified catalog for the redshifts of thousands of brand-new galaxies.

Methods

To verify the redshifts of our sources, the team visually inspected the spectral data and compared suggested redshift solutions from the automatic codes msaexp, LiMe, MARZ, Cigale, and BAGPIPES. Using PRISMATIC's flagging tool, we identified common spectral features such as [OIII]+Hβ, Hα, Lyα, Paα, the Lyman break, and the Balmer break. We can confidently estimate redshift when [OIII]+Hβ, Hα, or Paα are prominent. For each spectrum, our team assigned a confidence flag to each selected redshift, with four being the highest, indicating a secure redshift with several apparent features, and zero being the lowest, indicating that the redshift was unknown due to the absence of any detectable features. Additionally, a flag of nine indicated that the spectrum had a single high-confidence feature used for estimating the redshift.

Figure 2. Easily Identifiable Spectra from Galaxy 9677 with Redshift 4.45

Figures and Results

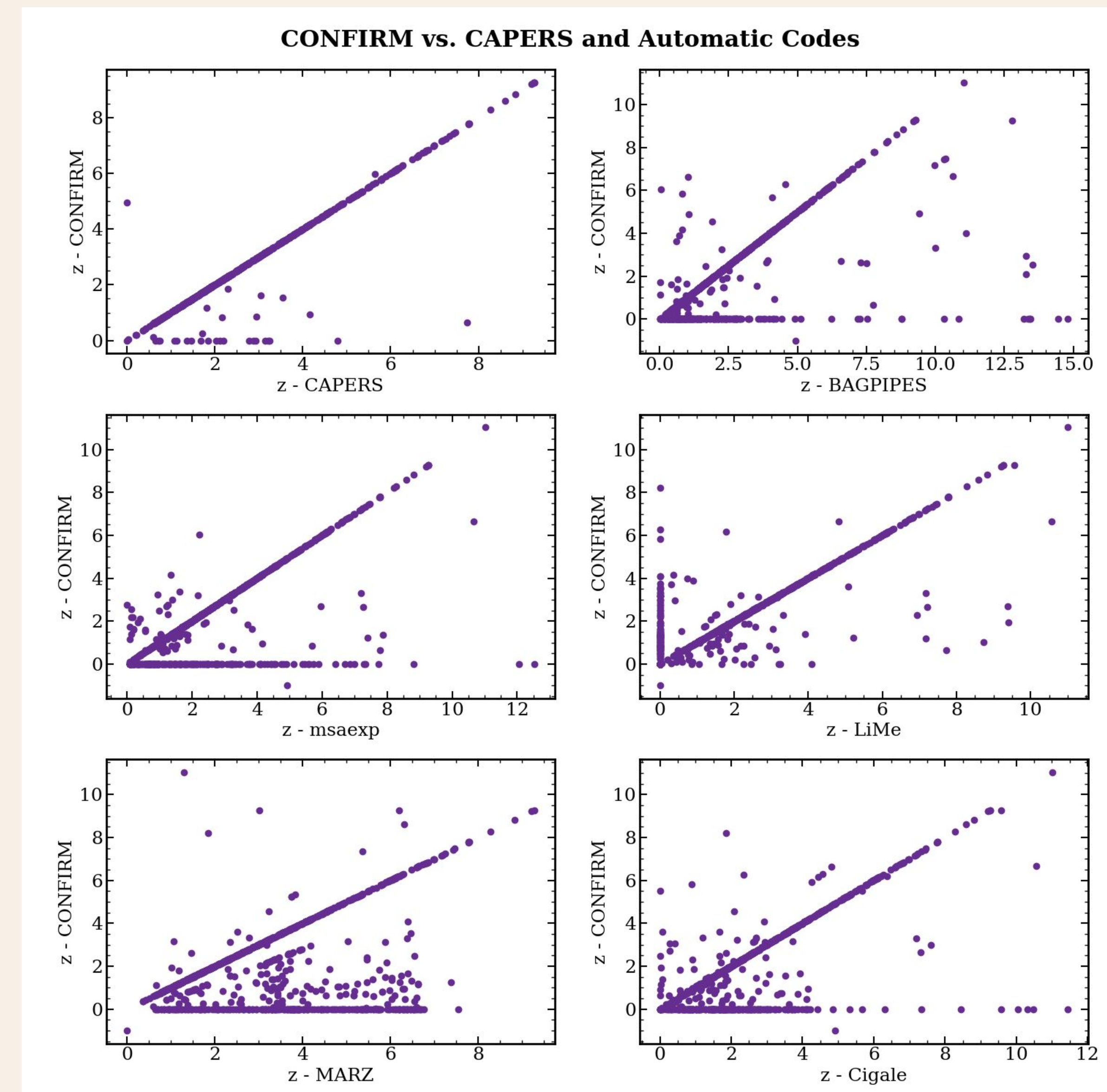

Figure 3. A comparison of the redshifts calculated by CONFIRM to those verified by the CAPERS team and those calculated by automated fitting codes.

Figure 3 highlights the inherent biases of the automated fitting codes, as well as the biases of our team. For instance, the automated codes Cigale, msaexp, and MARZ typically estimate a redshift when CONFIRM finds none. This phenomenon is flipped for LiMe, where CONFIRM typically finds a redshift even when LiME reports none. Furthermore, Cigale and BAGPIPES occasionally yield unrealistically high redshifts, which CONFIRM often disregards. Similarly, MARZ often reports a high redshift, but CONFIRM often reports a redshift that is half of MARZ's calculated value.

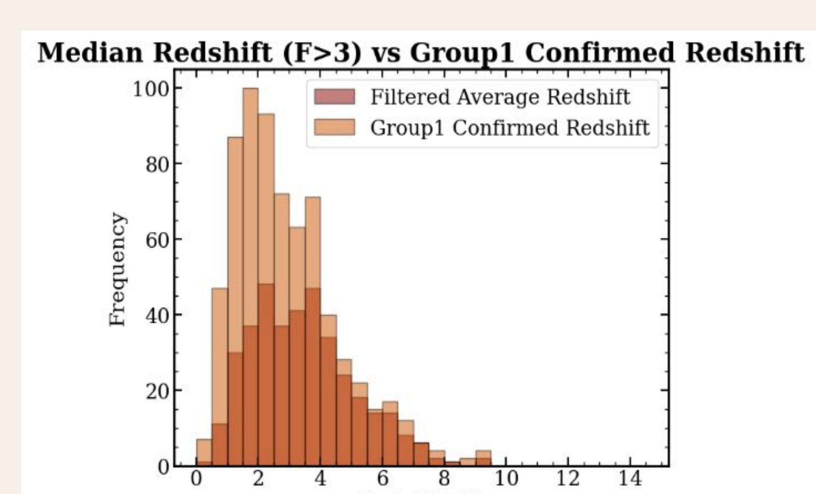

Figure 4a. The frequency of redshifts with flags over three calculated by CONFIRM and by CAPERS, scaled for visibility.

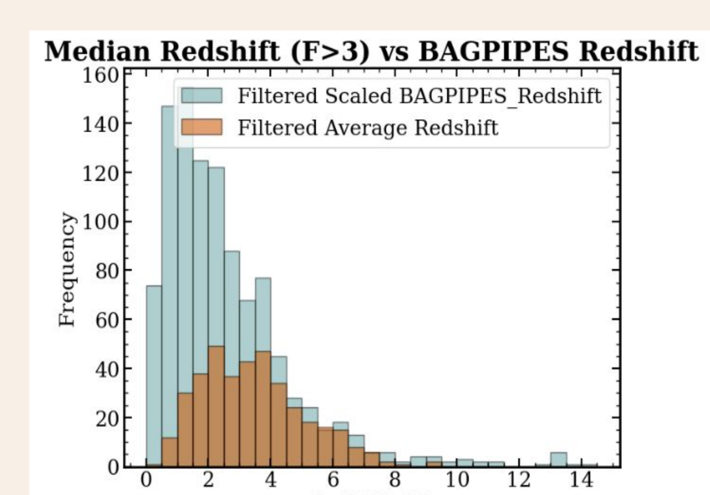

Figure 4b. The frequency of redshifts with flags over three calculated by CONFIRM and by the code BAGPIPES, scaled for visibility.

Figure 4a compares the filtered median redshifts with the high-confidence Group 1 Confirmed Redshift dataset from CAPERS, showing overall agreement but some differences—CAPERS appears more confident in lower redshifts (z < 4). Figure 4b compares median redshifts to BAGPIPES, which highlights key biases in BAGPIPES results, such as false positives at high redshifts (z > 6) and a lack of confidence in lower redshifts (z < 2). These discrepancies suggest that while BAGPIPES may overestimate some high-redshift sources, our dataset may underestimate lower-z values, warranting further refinement and validation.

Conclusion

We demonstrated that the PRISMATIC package is a useful tool for comparing various SED fitting models, and among the seven models tested, BAGPIPES and LiMe showed relatively accurate redshift estimations. However, each model also produced a significant number of results with notable discrepancies. For example, we found that the different redshift fitting codes obtain biases that can disrupt the accuracy of redshift values.

Our findings highlight that while well-known automated SED fitting tools are fast and convenient, cross-checking remains essential for achieving accurate redshift determinations. By comparing our cross-checked redshift values with confirmed values, we also evaluated the accuracy of CONFIRM and found that, although generally consistent, discrepancies remain with an RMS scatter of 0.56. This larger-than-expected value may be due to human error, such as variations in our group's methods of classifying redshifts.

We ensured that PRISMATIC is easily modifiable and reproducible by successfully analyzing over 1,000 spectra. A key outcome of this project is its scalability to larger datasets. Furthermore, it provides a valuable framework for astronomers aiming to determine redshifts using various SED fitting models. Looking ahead, we plan to confirm redshifts for an incoming dataset of CAPERS candidate high-redshift galaxy spectra. This ongoing effort will deepen our understanding of early-universe properties such as galaxy formation and evolution while also advancing techniques in astronomical data analysis.

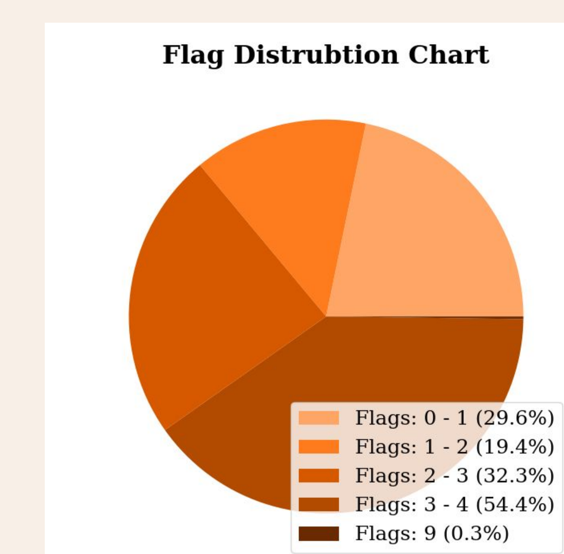

Figure 5. The distribution of flags (0, 1, 2, 3, 4, 9) by percentage.

Acknowledgments

Our team would like to thank CAPERS for providing the data necessary for the completion of this project. We would also like to acknowledge Sara Mascia and the PRISMATIC team for creating the program that allowed us to inspect the spectra and identify redshifts.

References

1. Dickinson, M., et al. (2024, March). The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS). Astrophysics Data System. https://ui.adsabs.harvard.edu/abs/2024jwst.prop.6368D.
2. Gaibler, V. Seeing Red. ESO Supernova. Retrieved from https://supernova.eso.org/exhibition/images/1111_DUM_2/.